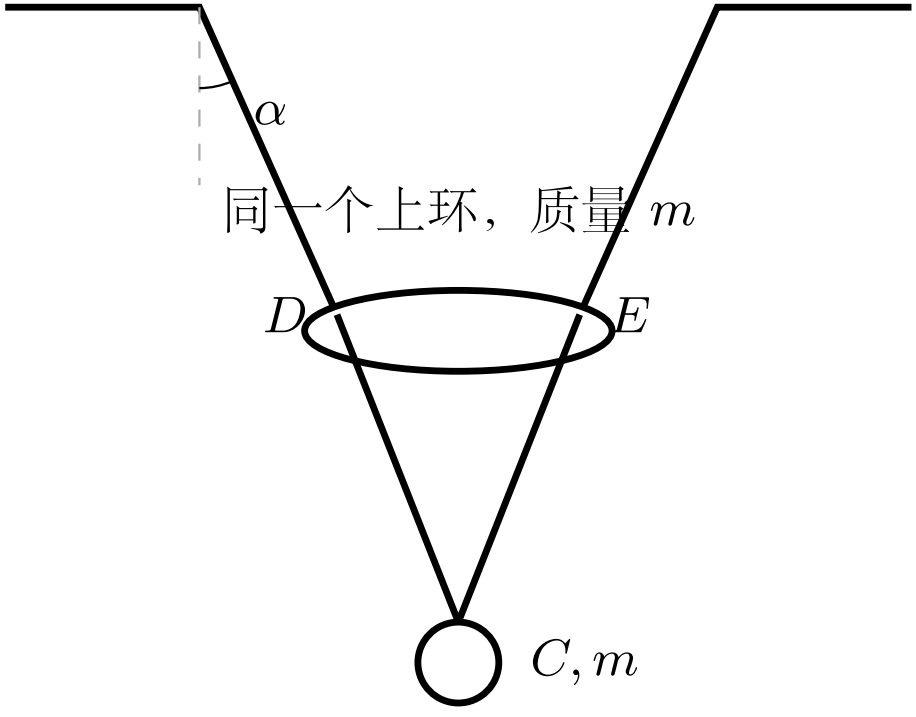
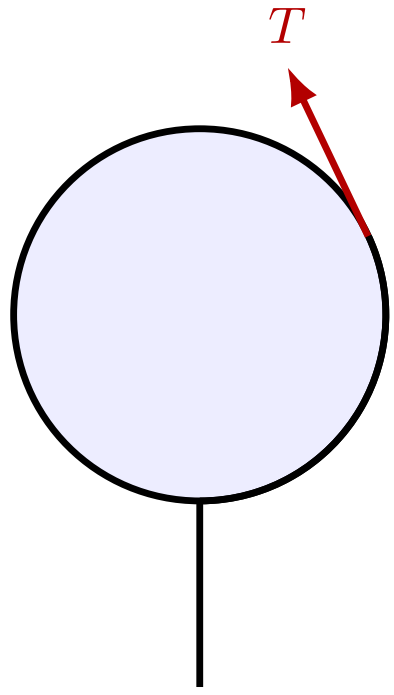
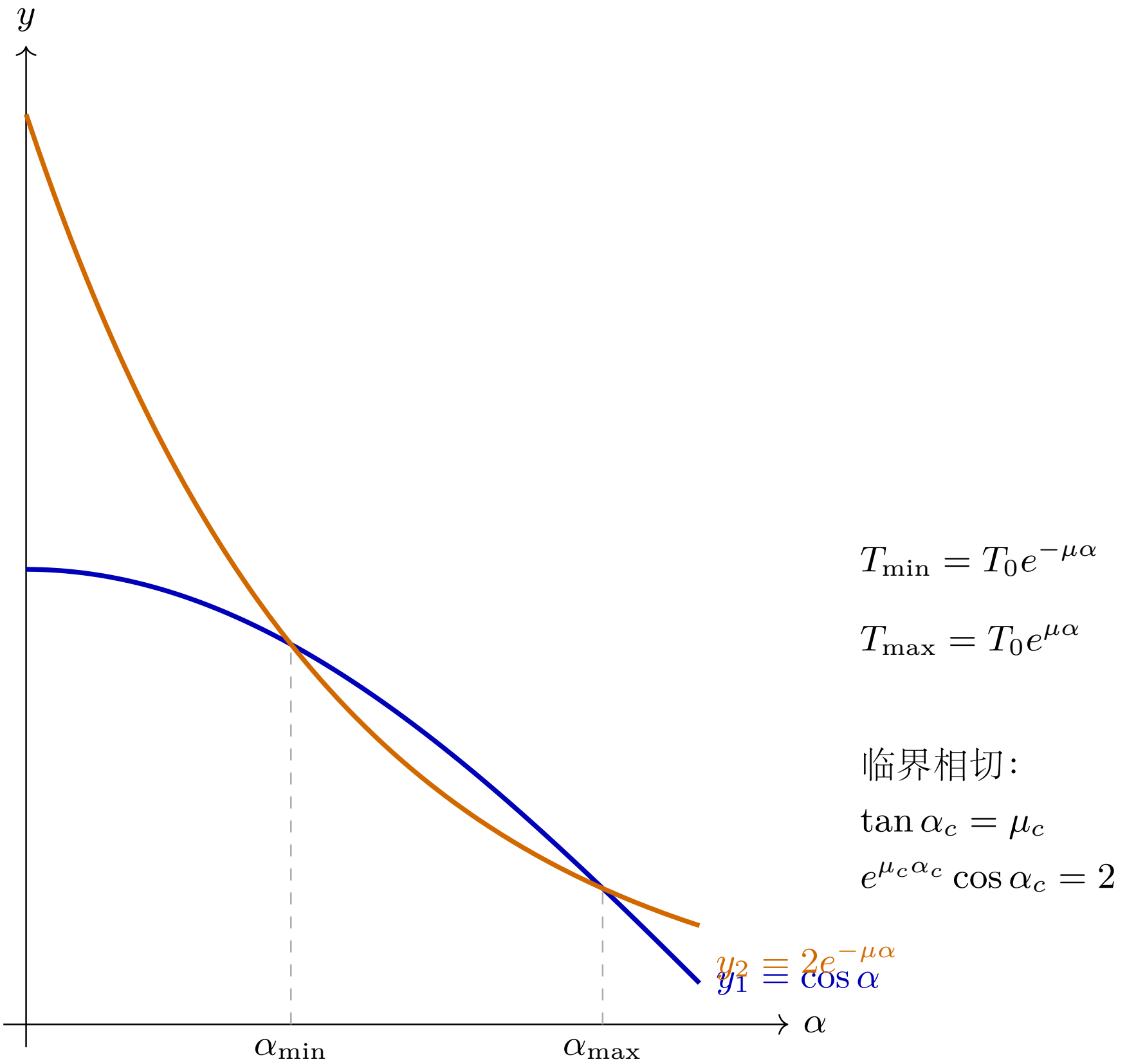


下环：$2T_0 = mg$

上环承受自重及两段绳的向下拉力



绕行角 α；$T/T_0 \leq e^{\mu\alpha}$



$$T_{\min} = T_0 e^{-\mu\alpha}$$

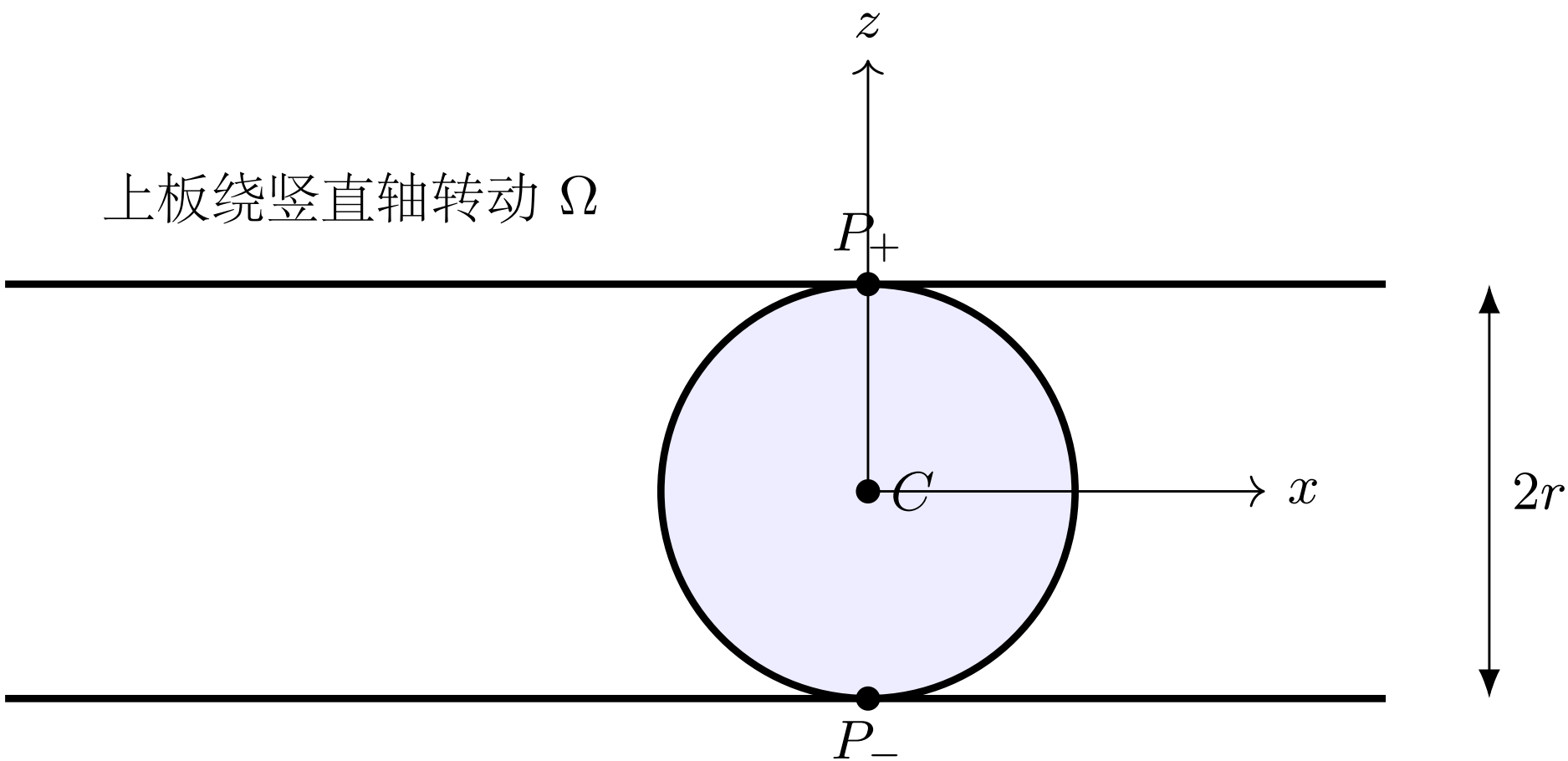
$$T_{\max} = T_0 e^{\mu\alpha}$$

临界相切：

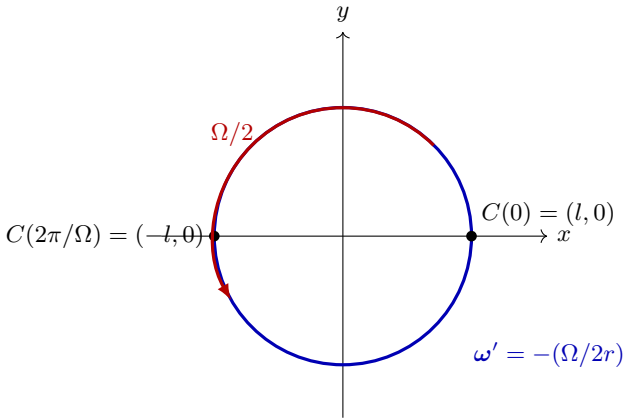
$$\tan\alpha_c = \mu_c$$

$$e^{\mu_c\alpha_c}\cos\alpha_c = 2$$

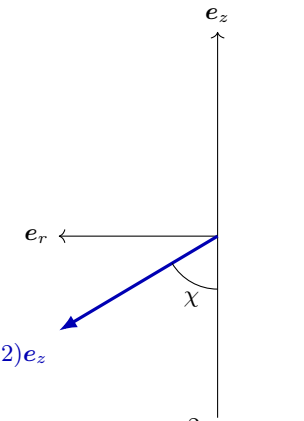
上环平衡：$2T\cos\alpha = mg + 2T_0 = 2mg$



侧视：上下接触点确在球面直径两端

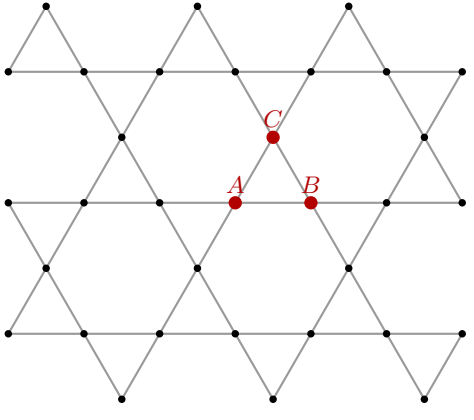


球心的半速圆周运动

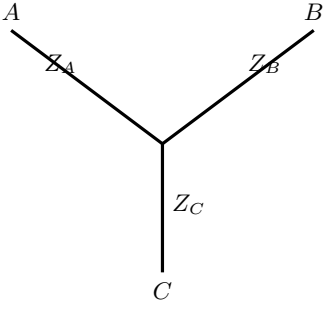


$$|\boldsymbol{\omega}'|\,\frac{2\pi}{\Omega}=2\pi n$$

$$\frac{r}{l}=\frac{1}{\sqrt{4n^2-1}}$$

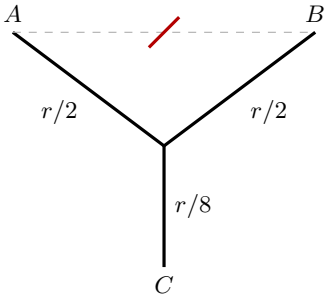


角点相接的三角网络：每节点四配位

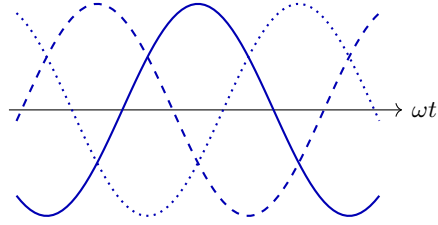


完整网络：$Z_A = Z_B = Z_C = r/4$

AB 直连电阻已断开



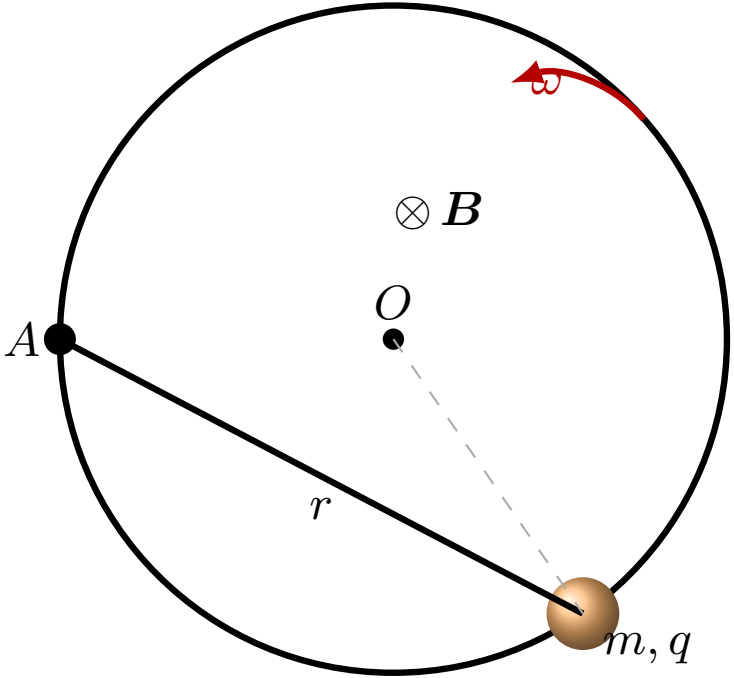
$R'_{AB} = r,\ \ R'_{AC} = R'_{BC} = 5r/8$



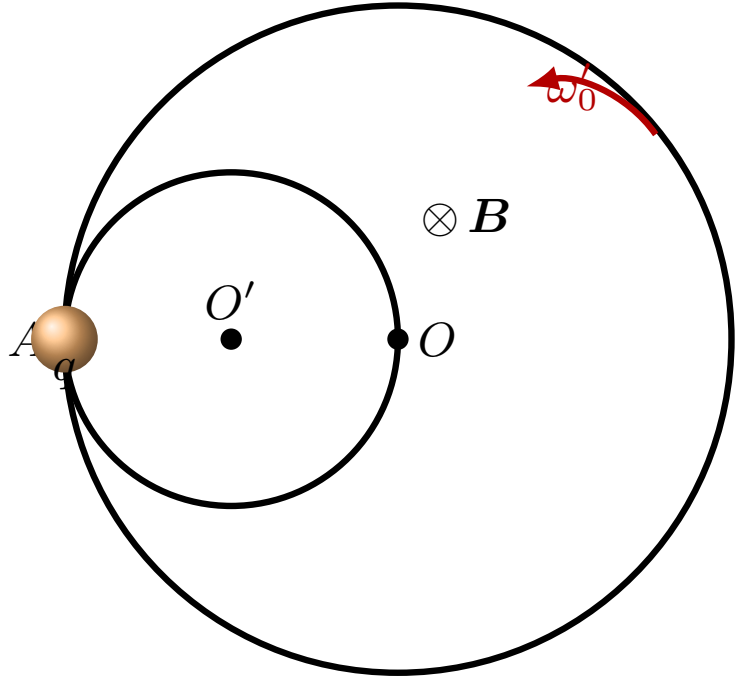
完整网络的三相驱动

$$u_A + u_B + u_C = 0$$

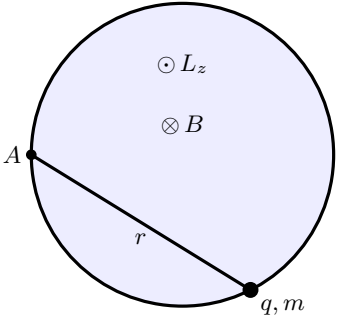
$$I_A = (4u_0/r)\cos\omega t$$



带电小珠在铰接绝缘环上滑动

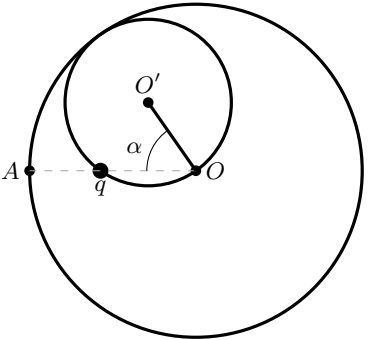


小环在大环内纯滚动



$L_z - \frac{1}{2}qBr^2 =$ 常量

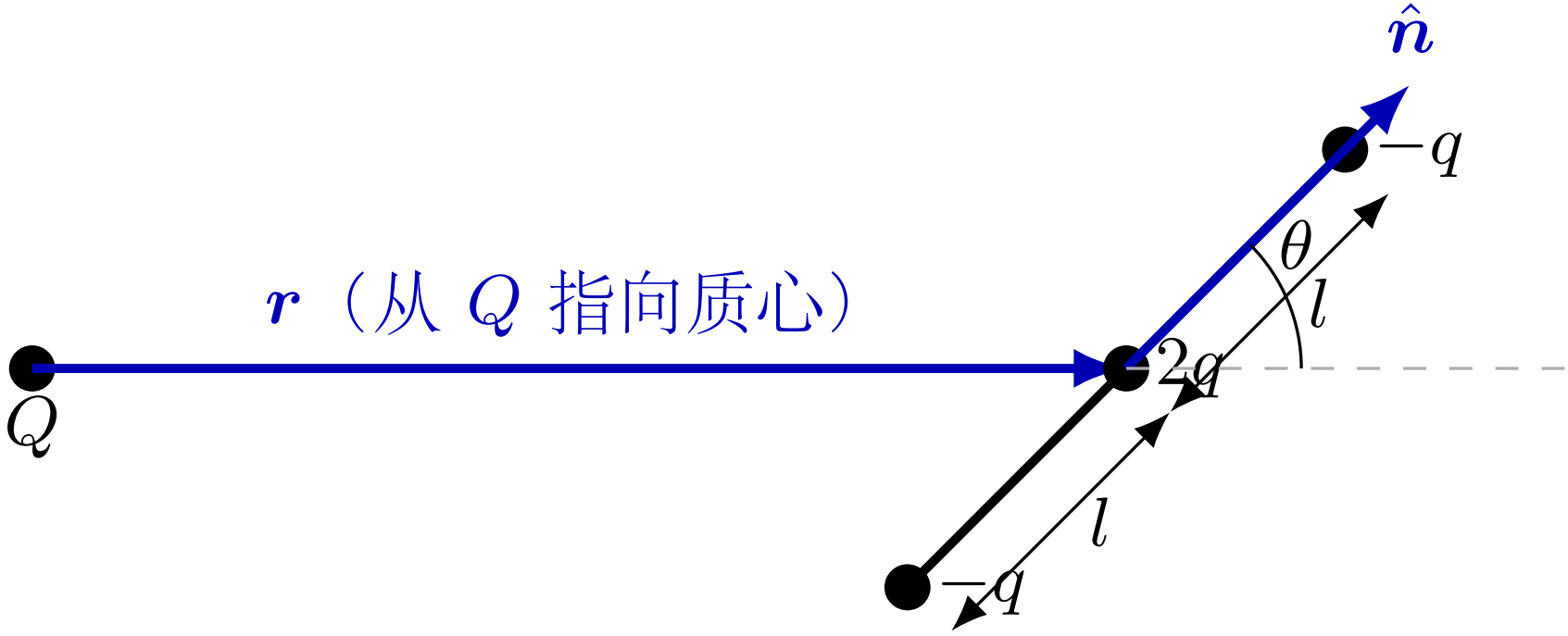
磁场不做功



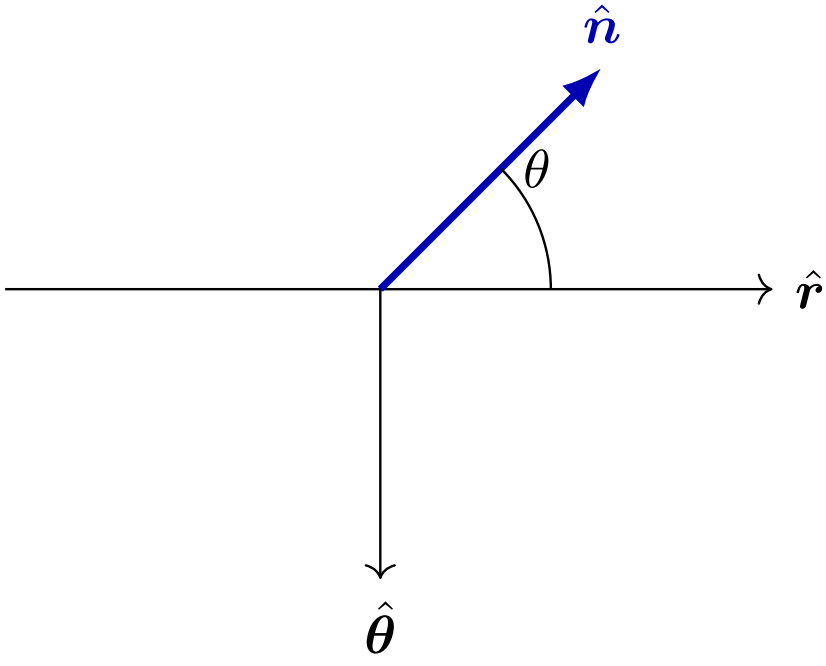
图取 $\alpha = 55°$；小环确在大环内

小珠在 AO 上：$r = R(1 - \cos\alpha)$

$\beta = 2\alpha$ 是相对接触半径的夹角



线性三电荷分子，总电荷与偶极矩均为零

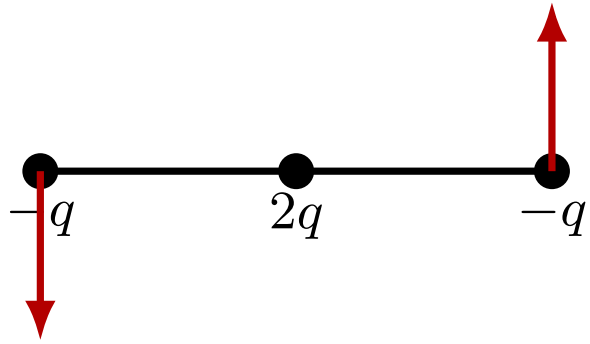


$$U = K(1 - 3\cos^2\theta)/r^3$$

$$F_r = 3K(1 - 3\cos^2\theta)/r^4$$

$$F_\theta = -6K\sin\theta\cos\theta/r^4$$

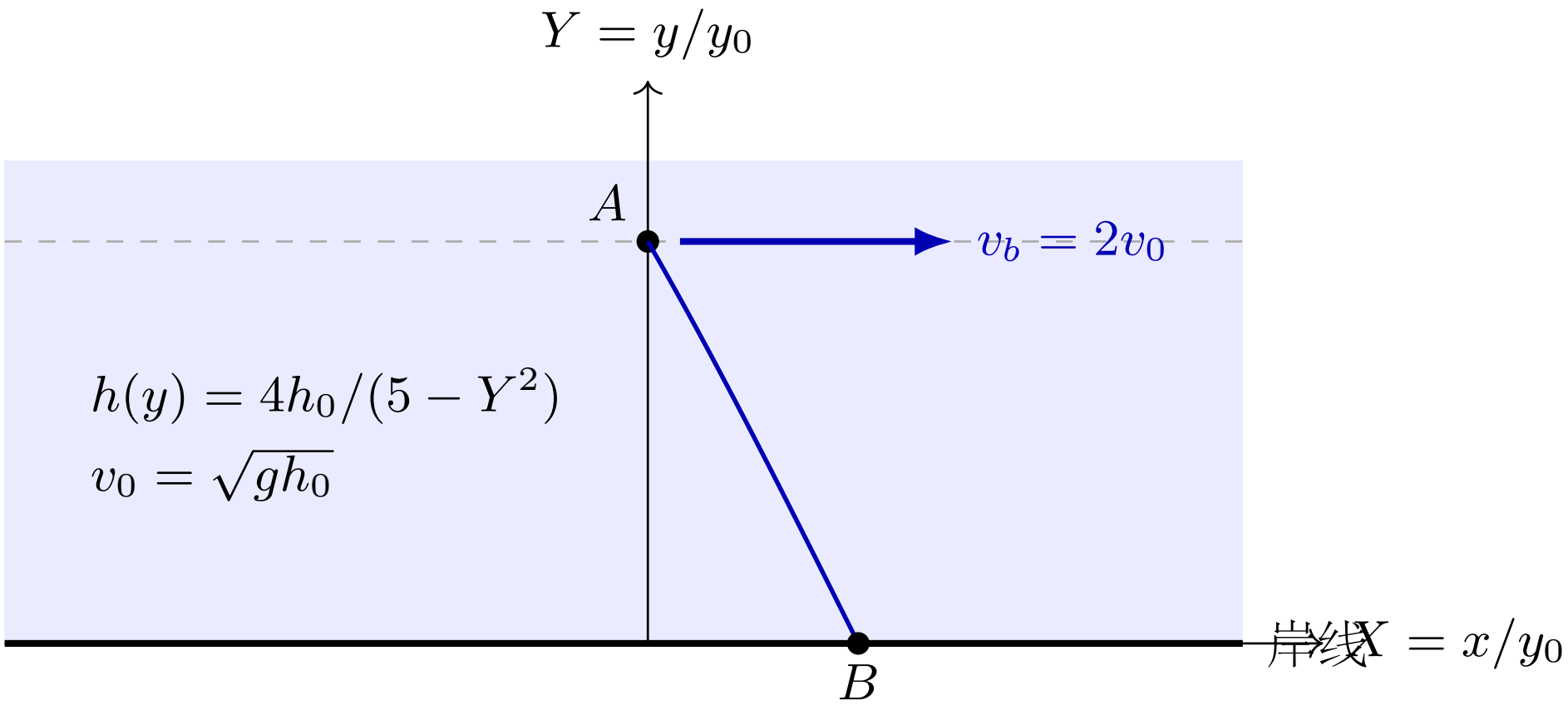
$\odot\, \boldsymbol{b}$（垂直分子）



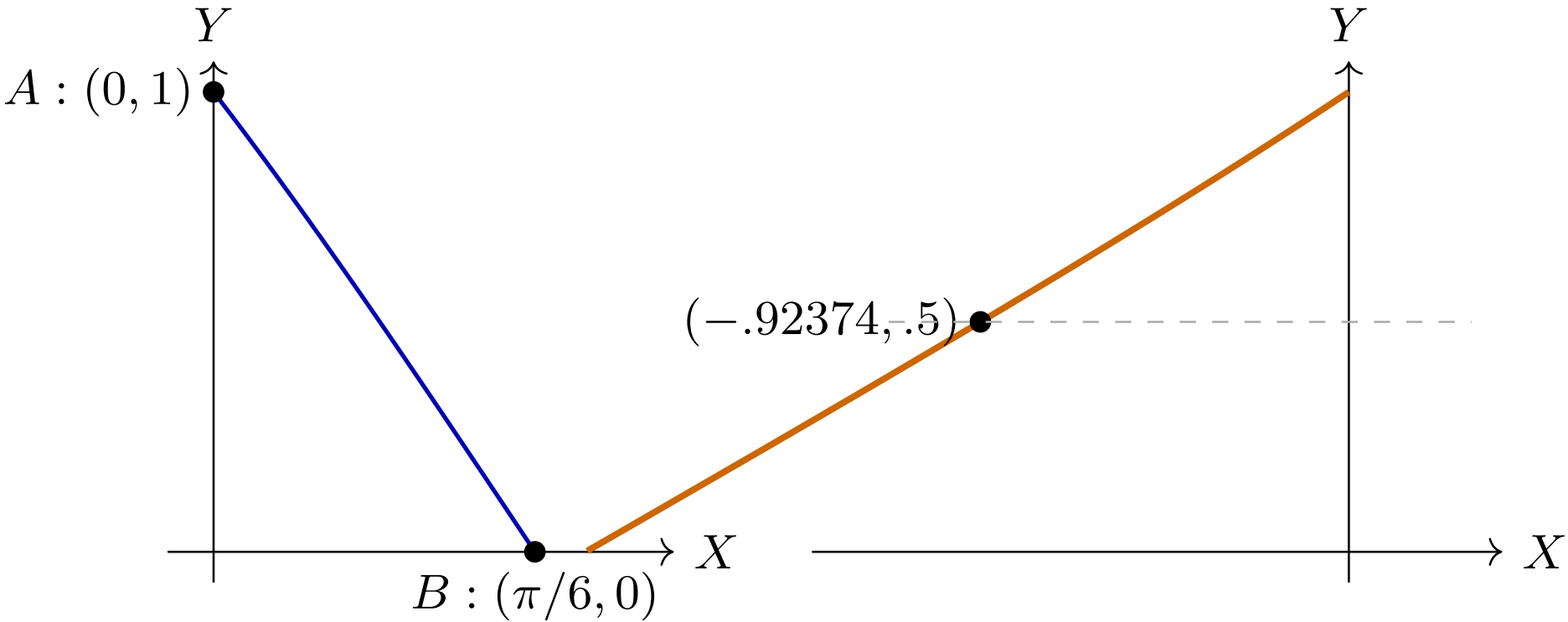
两端力相反、力矩同向

$$\boldsymbol{\tau} = ql^2[\boldsymbol{b} - \hat{\boldsymbol{n}}(\hat{\boldsymbol{n}} \cdot \boldsymbol{b})]$$

$\boldsymbol{b} \parallel \hat{\boldsymbol{n}}$ 时各端电场为零

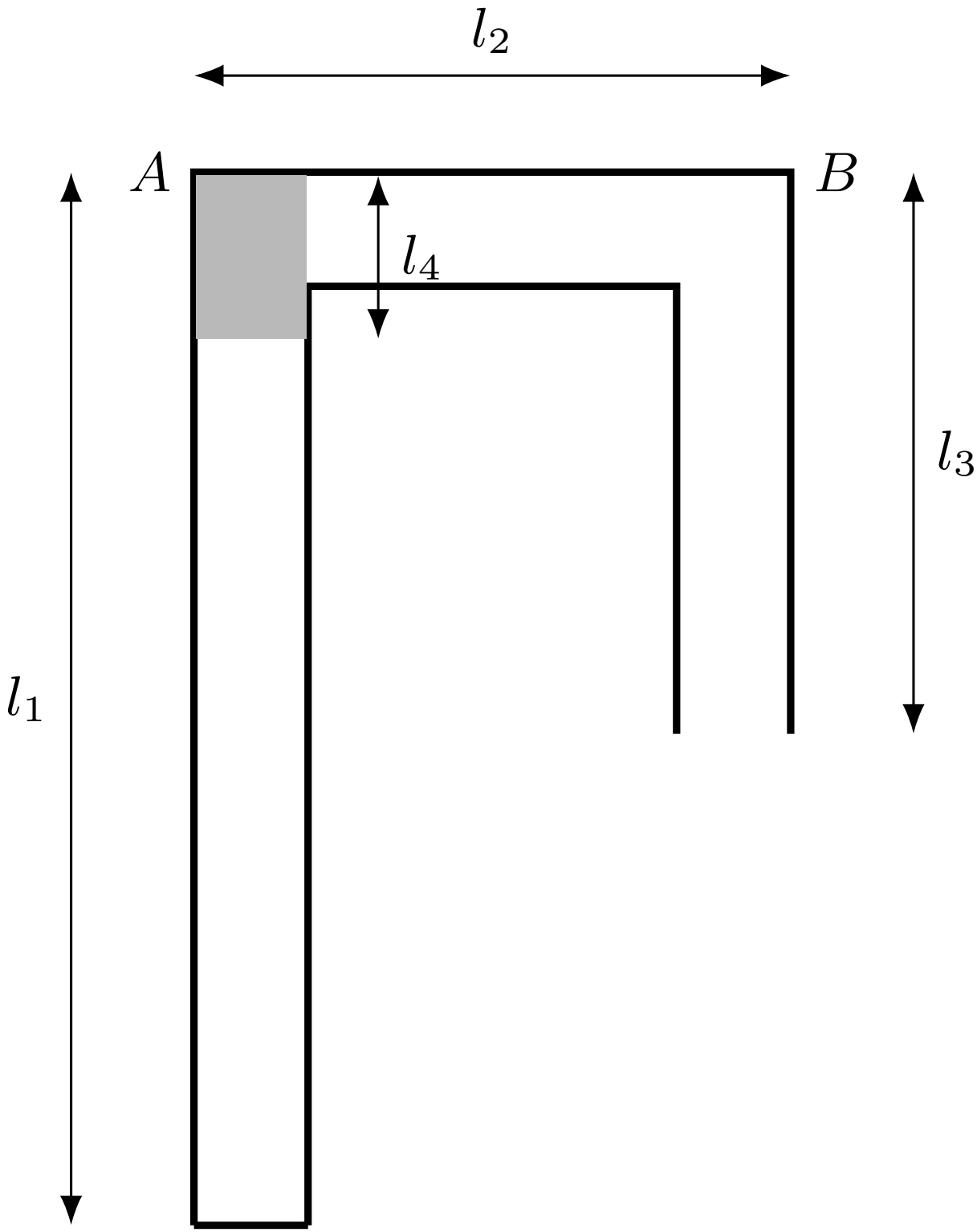


射线仅画 A 到岸线 B 的有效段



射线 $Y = 2\cos(X + \pi/3)$

同一时刻的波前，不是单条射线



$l_1 = 100\,\mathrm{cm}$

$l_2 = 5\,\mathrm{cm}$

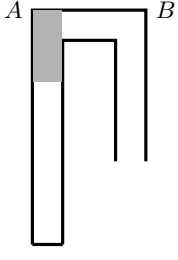
$l_3 = 50\,\mathrm{cm}$

$l_4 = 10\,\mathrm{cm}$

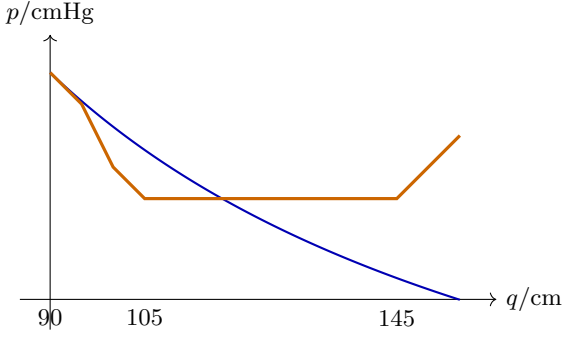
左端封闭

右端通大气

灰色为水银柱



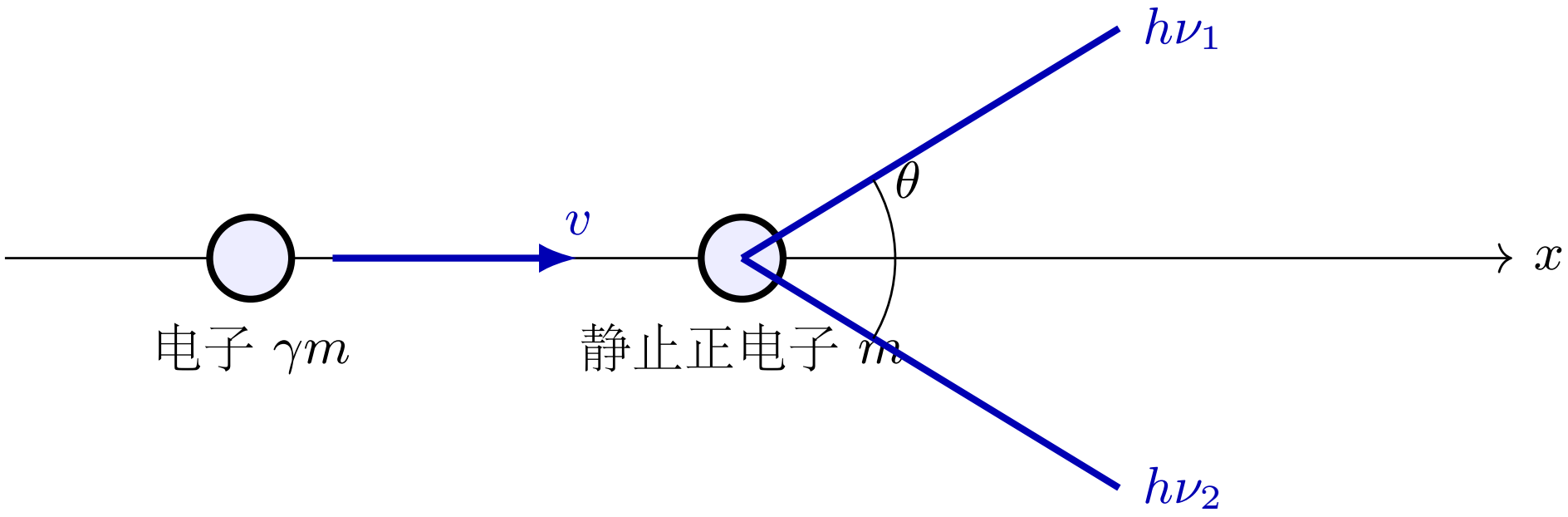
初态 $q_0 = 90$ cm，$p_i = 86$ cmHg



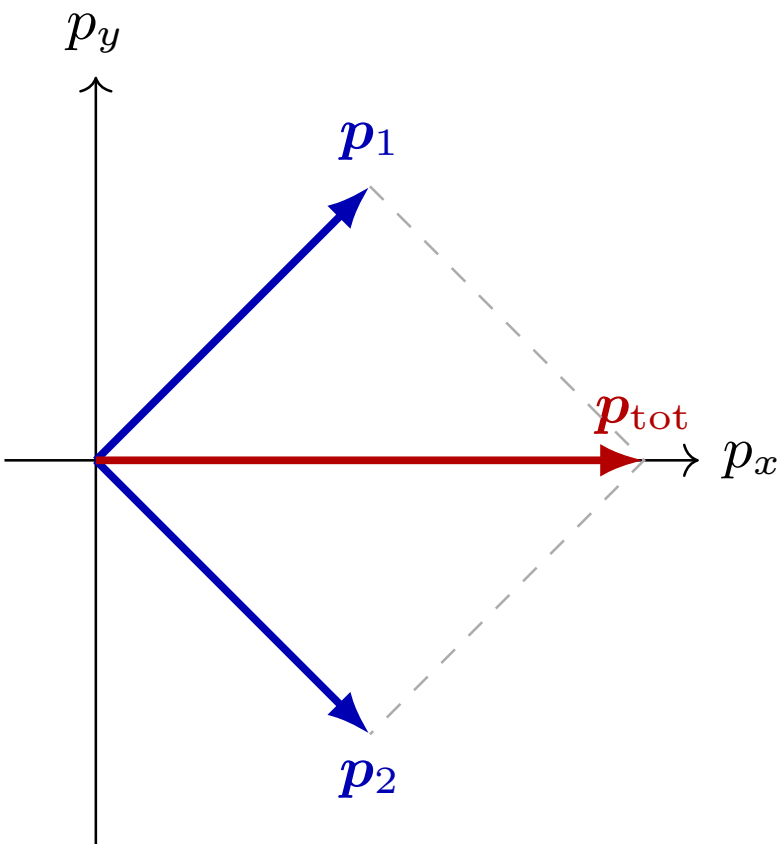
橙：实际分段平衡压力 p_{eq}

蓝：$p_g = 7740/q$，$T = 300$ K

正向初扰动失稳；弱阻尼后停在稳定交点



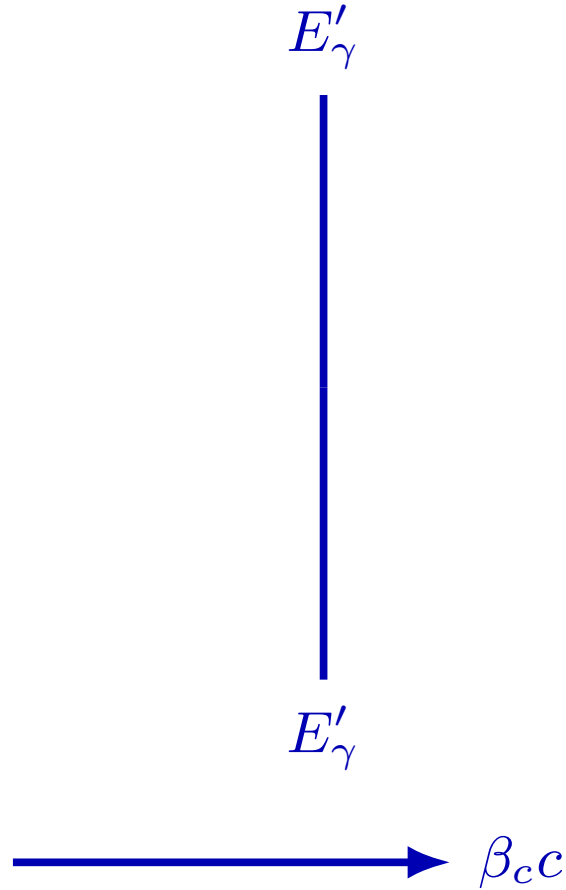
出射光子总动量沿入射电子方向



示例 $\gamma = 3$，等能光子夹角 $90°$

$$\boldsymbol{p}_1 + \boldsymbol{p}_2 = \boldsymbol{p}_{\mathrm{tot}}$$

质心系两光子等能、反向



$$\theta_{\min} = 2\arccos\sqrt{(\gamma-1)/(\gamma+1)}$$